

Lecture 10

Chapter 24
Electric Potential
Math Problems

❖ Potential energy difference

$$\Delta U = \frac{qq_0}{4\pi\epsilon_0} \left[\frac{1}{r_b} - \frac{1}{r_a} \right]$$

Where, $q_0 = \text{moving charge}$

$q = \text{source charge /charge at rest}$


$$\Delta V = \frac{\Delta U}{q_0}$$

❖ Potential difference

$$\Delta V = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_b} - \frac{1}{r_a} \right]$$

❖ Potential energy at a point

$$U = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r}$$

Where, $q_0 = \text{moving charge}$

$q = \text{source charge /charge at rest}$


$$V = \frac{U}{q_0}$$

❖ Potential at a point

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

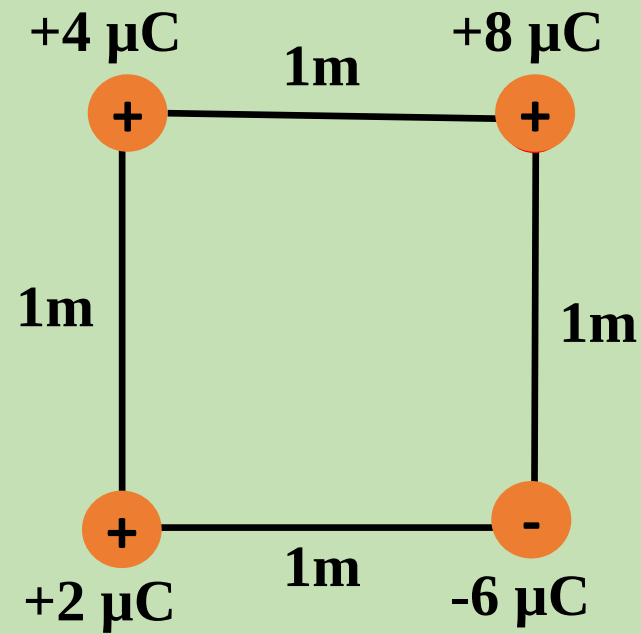
❖ Potential at a point for a group of charges

$$V = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1} + \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_2} + \frac{1}{4\pi\epsilon_0} \frac{q_3}{r_3} + \dots$$

Q1. The figure shows four charged particles held in fixed positions by forces that are not shown. What is the electric potential energy of this system of charges?

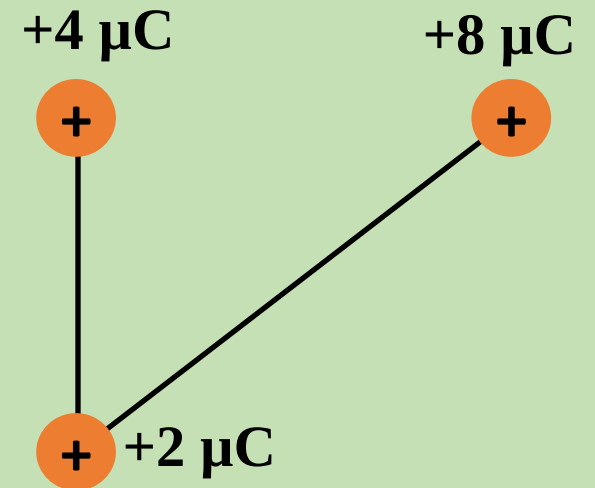
Ans:

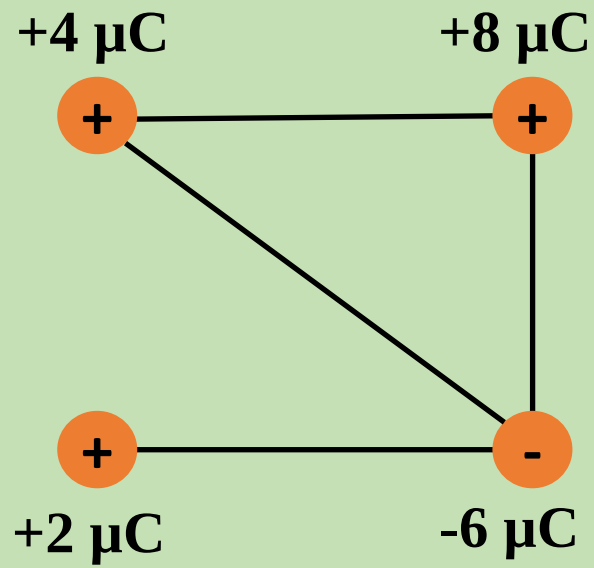
$$\begin{array}{c}
 \text{+} \\
 \text{+4 } \mu\text{C}
 \end{array}
 U_1 = 0$$



$$\begin{array}{c}
 \text{+4 } \mu\text{C} \qquad \qquad \text{+8 } \mu\text{C} \\
 \text{+} \text{-----} \text{+}
 \end{array}
 U_2 = 9 \times 10^9 \times \frac{4 \times 10^{-6} \times 8 \times 10^{-6}}{1}$$

$$U_3 = 9 \times 10^9 \times \frac{4 \times 10^{-6} \times 2 \times 10^{-6}}{1} + 9 \times 10^9 \times \frac{8 \times 10^{-6} \times 2 \times 10^{-6}}{\sqrt{2}}$$





$$U_4 = -9 \times 10^9 \times \frac{8 \times 10^{-6} \times 6 \times 10^{-6}}{1} - 9 \times 10^9 \times \frac{2 \times 10^{-6} \times 6 \times 10^{-6}}{1} - 9 \times 10^9 \times \frac{4 \times 10^{-6} \times 6 \times 10^{-6}}{\sqrt{2}}$$

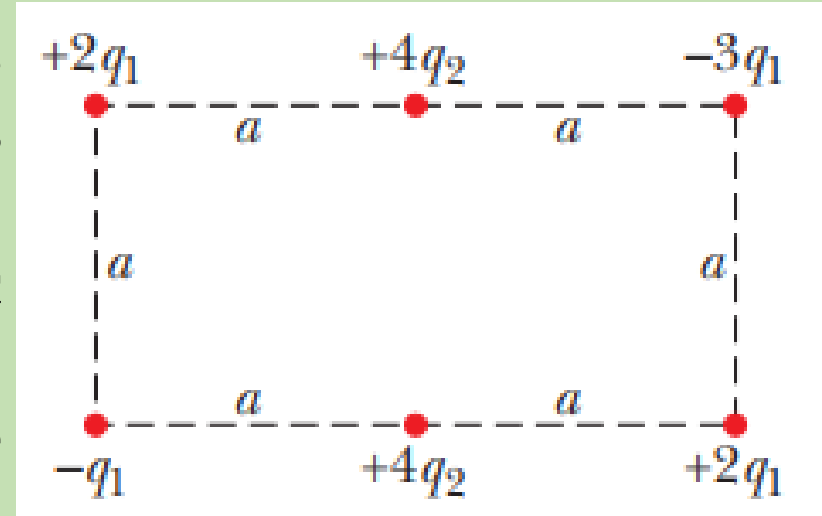
$$U = U_1 + U_2 + U_3 + U_4$$

Q2. Three charges, $4\mu\text{C}$, $-6\mu\text{C}$, and $8\mu\text{C}$ are placed at the corner of a square with sides equal to **1.0 m**. How much work is required to do so? (What is the potential energy of the system?)

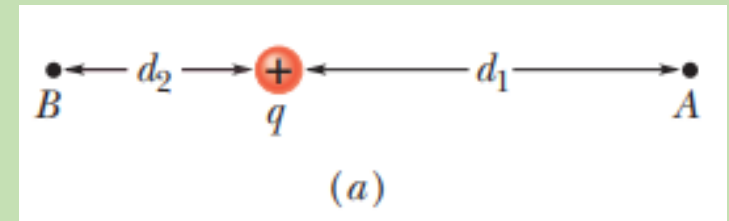
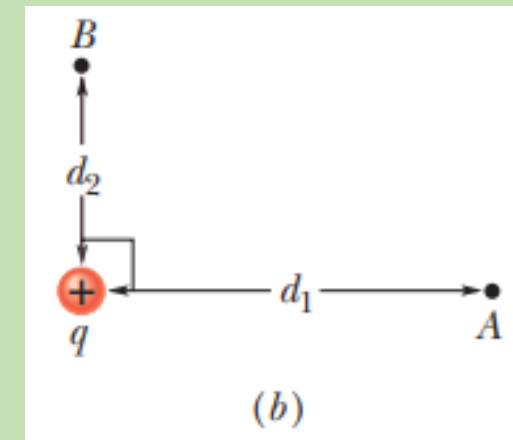
Do it now

Q3. Figure shows a rectangular array of charged particles fixed in place, with distance $a = 39.0 \text{ cm}$ and the charges shown as integer multiples of $q_1 = 3.40 \text{ pC}$ and $q_2 = 6.00 \text{ pC}$. With $V = 0$ at infinity, what is the net electric potential at the rectangle's center?

(Hint: Thoughtful examination of the arrangement can reduce the calculation.)



Q4. Consider a particle with charge $q = 1.0 \mu\text{C}$, point A at distance $d_1 = 2.0 \text{ m}$ from q , and point B at distance $d_2 = 1.0 \text{ m}$. (a) If A and B are diametrically opposite each other, as in Figure (a), what is the electric potential difference $V_A - V_B$? (b) What is that electric potential difference if A and B are located as in Figure (b)?



Q5. Two large, parallel, conducting plates are **12cm** apart and have charges of equal magnitude and opposite signs on their facing surfaces. An electric force of **$3.9 \times 10^{-15}\text{N}$** acts on an electron placed anywhere between the two plates. (Neglect fringing.) (a) Find the electric field at the position of the electron. (b) What is the potential difference between the plates?

Q6. A uniform electric field of 250 NC^{-1} is directed to the positive x -direction. A $12.0 \text{ }\mu\text{C}$ charge is placed from $x = 2.0 \text{ m}$ to the origin. How much work is required and what is the potential difference between the two points?

Thank You